

Frontiers in FinTech: Multimodal Foundation Models for Financial Reporting and Decision Science

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Submission Deadline: September 14, 2026

Abstract

The proliferation of unstructured, heterogeneous financial data-spanning PDF research reports, Excel-format financial statements, image-embedded market charts, and scanned policy documents poses a fundamental challenge for accounting information systems (AIS). Existing single-modality extraction pipelines and rule-based analytical tools fail to capture the full informational content embedded across these diverse formats, creating bottlenecks in financial statement analysis, audit evidence corroboration, and investment decision support. This study introduces **Frontiers in FinTech**, a multimodal large language model (MLLM) system that integrates Vision-Language Model (VLM) technology with domain-specific financial reasoning to deliver end-to-end intelligent financial analysis. We make three contributions to the AIS and accounting technology literature. First, we develop a Multimodal Financial Document Intelligence framework that leverages VLM encoders to simultaneously parse text, tables, and images from heterogeneous accounting documents, achieving unified structured extraction across format modalities and incorporating an automated cross-modal consistency validator that mirrors audit evidence corroboration procedures. Second, we introduce a Domain-Adaptive Two-Stage Training Protocol in which the LLM is pre-trained on large-scale public financial corpora (annual reports, accounting standards, research reports, market data) and subsequently fine-tuned on institution-specific investment data, enabling the model to master valuation methodologies (discounted cash flow, price-to-earnings, price-to-book, price-to-sales) and audit risk assessment frameworks with documented superiority over zero-shot and single-stage baselines. Third, we design a Natural Language Query-Driven Decision Pipeline that translates ambiguous user queries into executable financial analysis workflows, integrating portfolio optimization based on Modern Portfolio Theory, real-time risk monitoring, and multi-turn dialogue refinement. Validation across 200 listed companies demonstrates a 19 percent reduction in valuation error relative to the strongest published baseline, while a controlled user study with 48 accounting and investment professionals shows a 51 percent average reduction in task completion time. Implications are drawn for audit automation, financial reporting quality, and the democratization of expert-level financial analysis through AIS.

Keywords: Multimodal Large Language Model (MLLM); Accounting Information Systems (AIS); Domain-Adaptive Training (DAT); Intelligent Financial Analysis (IFA)

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1. Introduction

Accounting information systems (AIS) have long served as the backbone of corporate financial governance, enabling the collection, storage, processing, and reporting of financial data (?). Yet the exponential growth of financial data in both volume and variety over the past decade has outpaced the analytical capacity of traditional rule-based AIS architectures. Today, financial analysts, auditors, and investment professionals must synthesize information drawn from fundamentally different data formats: narrative management discussion sections in PDF annual reports, numerical income statements and balance sheets in Excel workbooks, technical indicators embedded in JPEG market-trend charts, and regulatory guidance distributed as scanned policy documents.

The inadequacy of existing systems in handling such heterogeneity carries measurable costs. ? estimate that financial analysts spend over 40 percent of their working hours on manual data collection and format conversion tasks that precede actual analysis. Auditors face growing challenges in verifying information consistency across disclosures that combine textual management discussion and analysis (MD&A), tabular footnotes, and graphical disclosures within a single filing (?). Meanwhile, investment research departments at institutional investors increasingly struggle to process the velocity of multi-format data generated by real-time market events, regulatory filings, and macro-policy announcements (?).

Recent advances in large language models (LLMs)—particularly Vision-Language Models (VLMs) (??)—offer a transformative opportunity to address these limitations. VLMs can jointly encode textual and visual information, enabling systems that understand the semantic content of a bar chart, parse a multi-column financial table, and reason over a scanned policy memo within a single unified inference pipeline. However, general-purpose VLMs lack the domain-specific financial reasoning necessary for professional-grade accounting applications. Without exposure to accounting standards (GAAP/IFRS), valuation conventions, and sector-specific financial logic, general LLMs produce hallucinated financial analyses or misapply valuation multiples (?).

This paper presents **Frontiers in FinTech**—a purpose-built MLLM system that closes this gap through three tightly integrated innovations. The system is designed to serve financial institutions, investment management firms, corporate accounting departments, auditing practices, and sophisticated individual investors who require timely, accurate, and explainable financial analysis at scale.

1.1 Research Questions

This study addresses three research questions:

RQ1. Can a VLM-based financial document intelligence framework extract accounting-relevant information from heterogeneous document formats (PDF, Excel, chart images, scanned policy documents) with accuracy comparable to credentialed human analysts?

RQ2. Does a two-stage training protocol (large-scale financial pre-training followed by institution-specific fine-tuning) enable an LLM to master professional accounting valuation methodologies more effectively than single-stage approaches?

RQ3. Does a natural language query-driven interface integrating portfolio theory, risk assessment, and multi-turn dialogue improve the quality and efficiency of investment decision-making for accounting professionals?

1.2 Contributions

This study makes the following contributions:

1. We design a **Multimodal Financial Document Intelligence** framework (Section ??) that unifies extraction from text, image, and tabular financial data, and incorporates a cross-modal consistency validator that automates a class of audit evidence corroboration procedures.
2. We introduce a **Domain-Adaptive Two-Stage LLM Training Protocol** (Section ??) that encodes financial accounting knowledge—including DCF, P/E, P/B, and P/S valuation models, MACD/RSI technical indicators, and GAAP/IFRS reporting standards—into a deployable MLLM, with documented superiority over zero-shot and single-stage baselines.
3. We propose a **Natural Language Query-Driven Decision Pipeline** (Section ??) that integrates Modern Portfolio Theory (MPT), real-time risk monitoring, and interactive multi-turn dialogue, lowering the professional barrier to institutional-quality financial analysis.
4. We discuss **implications for accounting practice** (Section ??), including audit automation, earnings quality assessment, and the evolving role of AIS in intelligent investment governance.

The remainder proceeds as follows. Section ?? reviews related work. Sections ??–?? present the three core innovations. Section ?? describes the integrated system architecture. Section ?? presents experimental results. Section ?? discusses implications. Sections ?? and ?? address limitations and conclusions.

2. Related Work

2.1 Accounting Information Systems and Data Heterogeneity

AIS have evolved from early transaction processing systems to enterprise resource planning (ERP) platforms and, most recently, to data analytics ecosystems (?). A persistent limitation of classical AIS architectures is their reliance on structured, machine-readable data; the growing preponderance of unstructured and semi-structured financial information has emerged as a central research concern (?).

? demonstrate that quantitative signals embedded in non-standard locations—such as narrative footnotes or graphical disclosures—carry incremental information content beyond structured financial statements. ? show that MD&A sentiment and forward-looking language predict future earnings revisions, underscoring the informational value of narrative content that rule-based systems routinely discard. These findings motivate a systems approach that treats all modalities of financial communication as first-class inputs to AIS.

2.2 Large Language Models in Finance and Accounting

The application of LLMs to finance and accounting has accelerated since the release of GPT-3 (?). ? introduced BloombergGPT, demonstrating that domain-specific pre-training on financial corpora yields significant improvements on financial NLP benchmarks. ? proposed FinGPT, an open-source financial LLM with a continuous data pipeline, and ? established the Financial Language Understanding Evaluation (FLUE) benchmark.

Within accounting specifically, ? survey AI applications across audit, tax, and management accounting, identifying LLM-based document review and anomaly detection as high-impact opportunities. ? examines how machine learning can automate aspects of audit evidence collection. ? find that ChatGPT sentiment scores derived from news headlines predict next-day stock returns, suggesting that LLM-derived signals carry genuine economic content.

A critical gap in the literature is the absence of systems that (a) handle multi-format, multi-modal financial data in a unified architecture, (b) encode professional-grade valuation and accounting knowledge through principled domain adaptation, and (c) surface that knowledge through a natural language interface designed for non-programmer accounting professionals. Frontiers in FinTech addresses this tripartite gap.

2.3 Vision-Language Models and Document Intelligence

VLMs such as CLIP (?), BLIP-2 (?), and LLaVA (?) demonstrate impressive cross-modal understanding. In document intelligence, LayoutLM (?) and Donut (?) show strong performance on structured document parsing. However, these models were not designed for the semantic com-

plexity of financial documents, which combine domain-specific terminology, industry-standard formatting conventions, and multi-referential data relationships (?).

Frontiers in FinTech builds on this VLM foundation but extends it with (1) a financial-domain pre-training corpus, (2) a multi-format data integration layer handling PDF research reports, Excel financial tables, and image-embedded market charts, and (3) a structured output layer mapping extracted information to standardized accounting fields.

2.4 Portfolio Optimization and Quantitative Risk Management

The theoretical foundation for portfolio construction in Frontiers in FinTech rests on Modern Portfolio Theory (?). In practice, institutional investment systems combine MPT-based optimization with real-time risk metrics including portfolio volatility, maximum drawdown, and the Sharpe ratio (?). AI-augmented portfolio management has been studied by ? and ?, who show that deep learning-based allocation strategies can outperform classical benchmarks under certain market regimes. Our contribution integrates MPT-based portfolio optimization with LLM-generated fundamental analysis and scenario-based risk stress testing, delivered through a conversational interface accessible to accounting professionals without quantitative programming skills.

3. Multimodal Financial Document Intelligence

3.1 Motivation and Problem Formulation

Financial accounting practice generates an extraordinary diversity of document formats. A single investment due-diligence workflow may require processing: (a) a PDF industry research report with narrative analysis, embedded tables, and high-resolution market-share charts; (b) an Excel workbook containing audited income statements, balance sheets, and EBITDA bridge calculations; (c) JPEG-format candlestick charts with technical indicator overlays; and (d) scanned Word-format regulatory policy documents.

Let $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$ denote a corpus of financial documents drawn from a heterogeneous format space $\mathcal{F} = \{\text{PDF}, \text{XLSX}, \text{IMG}, \text{DOCX}, \dots\}$. The goal of the multimodal financial document intelligence module is to learn a mapping:

$$f_{\text{parse}} : \mathcal{D} \rightarrow \mathcal{S} \quad (1)$$

where $\mathcal{S}$ is a structured, semantically annotated financial data schema containing fields such as revenue R_t , net income NI_t , earnings per share EPS_t , price-to-earnings ratio PE_t , industry growth rate g , and policy impact scores π .

3.2 VLM-Based Multi-Format Encoder Architecture

The parsing module employs a Transformer-based VLM backbone with dual processing paths for visual and textual tokens (?), augmented with three financial-domain extensions.

3.2.1 Adaptive Format Pre-processor

Rather than converting all inputs to a single canonical format, Frontiers in FinTech employs format-specific pre-processing pipelines.

pdf research reports. A hierarchical layout parser segments documents into sections, tables, and figures using spatial heuristics, then feeds each segment to the VLM encoder with document-position embeddings.

excel financial statements. A spreadsheet structure parser identifies row and column hierarchies, merges cells, and reconstructs multi-level header structures before encoding.

image-based charts. A chart-type classifier (bar, line, candlestick, scatter) invokes the appropriate axis-extraction routine to recover numerical series from visual encodings.

policy documents. An OCR pipeline with domain-specific post-correction handles scanned materials, followed by clause-level segmentation for regulatory parsing.

3.2.2 Financial Field Extractor

After encoding, a cross-attention extraction head maps VLM representations to a predefined financial ontology containing over 120 standardized accounting fields aligned with GAAP and IFRS reporting categories. Key extracted fields include revenue (R_t), operating income ($EBIT_t$), net income (NI_t), total assets, total liabilities, free cash flow, capital expenditures, P/E ratio, P/B ratio, P/S ratio, industry revenue growth rate, dividend yield, and qualitative signals such as policy favorability scores and management tone indices.

3.2.3 Cross-Modal Consistency Validator

A distinguishing feature of Frontiers in FinTech’s parsing layer is an explicit cross-modal consistency validation step. When the same financial figure (e.g., 2024 annual revenue) appears in both a narrative PDF and an Excel table within the same corpus, the system automatically flags discrepancies exceeding a configurable materiality threshold ϵ_m :

$$\text{Alert} = \mathbf{1} \left[\left| \frac{v_{\text{PDF}} - v_{\text{XLSX}}}{v_{\text{XLSX}}} \right| > \epsilon_m \right] \quad (2)$$

This automated consistency check mirrors a core audit procedure—the agreement of financial figures across different disclosure channels— and represents a novel contribution to audit-support

AIS design.

3.3 Output: Standardized Financial Data Set

The parsing module produces a standardized data set $\hat{S}$ reporting: (i) structured numerical financial metrics per entity per period; (ii) source attribution (document type, page or sheet, bounding box coordinates) for every extracted figure, enabling audit-trail traceability; (iii) data quality scores per field (completeness, cross-document consistency, extraction confidence); and (iv) a human-readable data collection summary (e.g., “15 valid research reports and 30 market data series collected; data completeness 92 percent”).

3.4 Significance for Accounting Practice

This innovation addresses the *multi-format heterogeneity problem* that has persistently limited AIS capabilities. From an accounting perspective, the cross-modal consistency validator directly automates a class of audit procedures relating to the agreement of financial figures across disclosure channels. The source-attributed output additionally satisfies the evidentiary traceability requirements of audit standards (e.g., AS 1105), positioning Frontiers in FinTech as a complement to professional judgment in evidence-gathering contexts.

4. Domain-Adaptive Two-Stage LLM Training

4.1 Motivation: The Gap Between General AI and Accounting Expertise

General-purpose LLMs exhibit systematic deficiencies when applied to professional accounting tasks. Empirical tests conducted during Frontiers in FinTech’s development revealed that off-the-shelf GPT-class models (a) misapplied DCF discount rates to growth-stage versus mature-firm scenarios, (b) confused P/B ratios with P/S multiples for capital-intensive industries, (c) failed to recognize GAAP-specific disclosure conventions, and (d) generated overconfident valuations unsupported by stated assumptions. These failures reflect the mismatch between the broad distributional training of general LLMs and the convention-bound knowledge domain of financial accounting and investment analysis. Closing this gap requires not merely more data, but a structured curriculum that encodes professional accounting knowledge into the model’s representations.

4.2 Stage 1: Large-Scale Financial Domain Pre-Training

4.2.1 Pre-Training Corpus

The Stage 1 corpus $\mathcal{C}_1$ is assembled from publicly available financial data encompassing: financial statements (annual and quarterly reports for listed companies across major exchanges, spanning at least 10 fiscal years per company); equity research reports from major financial institutions; histor-

ical time-series data for price, volume, and technical indicators; regulatory and policy documents; and investment case studies. The corpus contains approximately 10^{10} tokens after deduplication and quality filtering, with multimodal coverage (text, tables, charts) to support the VLM backbone.

4.2.2 Pre-Training Objectives

Stage 1 employs a multi-task pre-training objective:

$$\mathcal{L}_{\text{pre}} = \lambda_1 \mathcal{L}_{\text{LM}} + \lambda_2 \mathcal{L}_{\text{ITM}} + \lambda_3 \mathcal{L}_{\text{ITG}} + \lambda_4 \mathcal{L}_{\text{fin}} \quad (3)$$

where $\mathcal{L}_{\text{LM}}$ is standard language modeling loss, $\mathcal{L}_{\text{ITM}}$ is image-text matching loss, $\mathcal{L}_{\text{ITG}}$ is image-to-text generation loss, and $\mathcal{L}_{\text{fin}}$ is a novel financial reasoning loss that penalizes valuation outputs inconsistent with disclosed financial metrics (e.g., a calculated P/E ratio that contradicts stated earnings per share).

4.3 Stage 2: Institution-Specific Fine-Tuning

4.3.1 Fine-Tuning Data and Protocol

Stage 2 adapts the pre-trained model to the specific investment logic, risk framework, and analytical style of the deploying institution. Fine-tuning data $\mathcal{C}_2$ may include internal investment memos and historical trade rationales, institution-specific risk scoring models and portfolio construction rules, proprietary research frameworks, and historical analyst feedback (reinforcement learning from human feedback, RLHF). Fine-tuning employs parameter-efficient techniques including Low-Rank Adaptation (LoRA) (?) and Prompt Tuning, preserving the broad financial knowledge from Stage 1 while efficiently acquiring institution-specific knowledge with minimal catastrophic forgetting.

4.3.2 Valuation Model Library

A key output of the two-stage training is the model’s mastery of a structured valuation model library comprising the methodologies shown in Table 1.

4.3.3 Automated Valuation Report Generation

Following valuation computation, the model generates a structured report containing: (i) core financial metric summary, (ii) valuation parameter assumptions with sensitivity analysis, (iii) target price estimate with confidence interval, (iv) valuation reasonableness check against peer multiples, and (v) an investment rating (Buy / Hold / Sell) with explicit risk factors flagged. The report is calibrated to comply with sell-side research report disclosure standards (e.g., CFA Institute Research Objectivity Standards), facilitating its use in regulated investment advisory contexts.

Table 1. Valuation Methodologies Encoded in Frontiers in FinTech’s Training Curriculum

Method	Description	Best Applicable Sector
DCF	Discounted cash flow; projects FCFF/FCFE and discounts at WACC or equity cost of capital	Capital-intensive; stable cash flows
P/E	Earnings multiple relative to sector and growth peers	Consumer, technology, financial services
P/B	Asset-based multiple; applicable when book value approximates fair value	Banking, real estate, insurance
P/S	Revenue multiple; used when earnings are negative or volatile	Early-stage growth, SaaS, biotech
EV/EBITDA	Enterprise value to operating earnings; capital-structure neutral	Cross-border M&A, leveraged buyouts
Residual Income	Accounting-based value; anchored on book value and abnormal earnings	Earnings-focused, regulated industries

Note: The model automatically selects the appropriate method based on entity type, life cycle stage, and data availability. Multiple methods are applied and triangulated.

4.4 Accounting-Theoretic Grounding

The two-stage training protocol reflects the accounting literature’s insight that professional judgment in financial analysis is built on a structured body of domain knowledge (?). By explicitly encoding GAAP/IFRS financial statement relationships (e.g., the clean surplus relation linking book value, earnings, and dividends) and valuation theory (e.g., the Gordon Growth Model as a special case of DCF) into the pre-training curriculum, Frontiers in FinTech ensures that its outputs are grounded in theoretically consistent accounting logic rather than statistical pattern-matching alone. This distinction is critical for auditability and regulatory acceptability of AI-generated financial analyses.

5. Natural Language Query-Driven Decision Pipeline

5.1 Motivation: Democratizing Expert-Level Financial Analysis

A persistent challenge in the adoption of advanced financial analytics systems is the requirement for specialized programming or query skills. Traditional analytical platforms (e.g., Bloomberg Terminal, FactSet) require users to learn platform-specific query languages, creating a high expertise barrier that restricts access to quantitatively trained professionals. This barrier exacerbates the information asymmetry between institutional and individual investors identified in the market microstructure literature (?) and limits the scope of AIS-supported decision-making within accounting departments. Frontiers in FinTech’s third innovation addresses this through a **Natural Language Query-Driven Decision Pipeline** that enables users to express analysis requirements in ordinary prose and receive institutionally calibrated outputs without programming knowledge.

5.2 Query Understanding and Intent Resolution

5.2.1 Multi-Level Query Parsing

The system processes user queries through a three-level intent resolution hierarchy.

task classification. The query is classified into one of five primary task categories: (a) data collection, (b) market trend analysis, (c) company valuation, (d) portfolio construction and risk management, and (e) investment question-and-answer and insight retrieval.

entity and parameter extraction. Named entities (company names, tickers, sectors, time periods), quantitative parameters (target yield, risk tolerance, investment horizon), and analytical constraints (valuation method preference, geographic scope) are extracted via a fine-tuned named entity recognition model.

ambiguity resolution. When the query contains underspecified parameters (e.g., “analyze the pharmaceutical sector” without specifying sub-sector or time window), the system initiates a structured clarification dialogue: *“Please confirm the sub-sector: innovative drugs / medical devices / healthcare services?”*, ensuring the downstream analysis is precisely scoped.

5.2.2 Automated Task DAG Construction

Once intent is resolved, the system generates a Directed Acyclic Graph (DAG) of analytical sub-tasks:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \quad (4)$$

where nodes $v \in \mathcal{V}$ represent atomic analysis operations (data fetch, VLM parse, metric compute, model run, report generate) and edges $e \in \mathcal{E}$ encode data dependencies. The DAG scheduler dispatches tasks to the appropriate system modules in topologically sorted order, enabling parallel execution of independent subtasks and full transparency into analysis progress.

5.3 Portfolio Construction and Risk Management Integration

5.3.1 MPT-Based Portfolio Optimizer

The portfolio construction module implements Markowitz mean-variance optimization (?):

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} \geq \mu^*, \mathbf{w}^\top \mathbf{1} = 1, w_i \geq 0 \quad \forall i \quad (5)$$

where $\mathbf{w}$ is the portfolio weight vector, Σ is the estimated return covariance matrix, and $\boldsymbol{\mu}$ is the expected return vector derived from Frontiers in FinTech’s valuation outputs. The user’s risk preferences (target yield μ^* , maximum volatility, investment horizon) are extracted from the natural language query and mapped to formal optimization constraints.

5.3.2 Real-Time Risk Monitoring

The risk management sub-module continuously computes a set of portfolio risk indicators:

$$\begin{aligned} \sigma_p &= \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}} \quad (\text{portfolio volatility}) \\ \text{MDD} &= \max_{t \leq s} \frac{V_t - V_s}{V_t} \quad (\text{maximum drawdown}) \\ \text{SR} &= \frac{\mu_p - r_f}{\sigma_p} \quad (\text{Sharpe ratio}) \end{aligned} \quad (6)$$

and identifies concentration risk (sector exposure exceeding 30 percent), sector rotation signals, and macro-economic risk flags (e.g., interest rate sensitivity for bond-heavy portfolios). When risk thresholds are breached, the system surfaces specific rebalancing recommendations in natural lan-

guage: “Increasing government bond allocation by 20 percent is recommended to reduce portfolio volatility from 18 percent to 14 percent.”

5.3.3 Dynamic Rebalancing via Query

Portfolio rebalancing is triggered through natural language queries. A user may issue: “Adjust portfolio allocation based on current market conditions,” and the system will automatically (a) fetch the latest market data, (b) recompute optimal weights under updated expectations, (c) generate a proposed rebalancing schedule respecting transaction cost constraints, and (d) explain the logic in plain language.

5.4 Multi-Turn Dialogue for Progressive Analysis Refinement

Frontiers in FinTech supports multi-turn dialogue to enable progressive deepening of analysis. In a representative example:

Turn 1 (User): “Analyze 2024 A-share liquor sector monthly trend and three-month forecast.”

Turn 1 (System): Generates sector trend report with MACD/RSI signals, support/resistance levels, and probability-weighted price scenarios.

Turn 2 (User): “Elaborate on capital flow patterns into the liquor sector by sub-industry.”

Turn 2 (System): Drills into inter-sub-industry capital rotation dynamics with granular flow data.

The multi-turn mechanism maintains a session context graph tracking resolved entities, prior outputs, and user preferences across turns, enabling co-reference resolution and incremental refinement without requiring re-specification of full analytical context.

5.5 Significance for AIS Research

The natural language query-driven decision pipeline addresses the *expertise barrier problem* in accounting and investment analysis. Rather than requiring users to learn the system’s query language, the system accommodates users’ natural expression patterns. This reconceptualization has important implications for accounting practice, particularly for non-quantitative professionals (financial controllers, internal auditors, CFOs) who require analytical outputs but lack the technical skills to operate traditional AIS query interfaces.

6. System Architecture

6.1 Three-Layer Architecture

The three innovations described in Sections ??–?? are integrated within a unified three-layer architecture: (1) the Multimodal Data Layer ingests and parses heterogeneous financial documents into standardized structured fields; (2) the Financial Intelligence Layer applies domain-adapted LLM reasoning to generate valuations, market analyses, risk assessments, and investment recommendations; and (3) the Interaction and Decision Layer manages the natural language interface, task DAG execution, portfolio optimization, and result delivery. Figure 1 illustrates the integrated architecture.

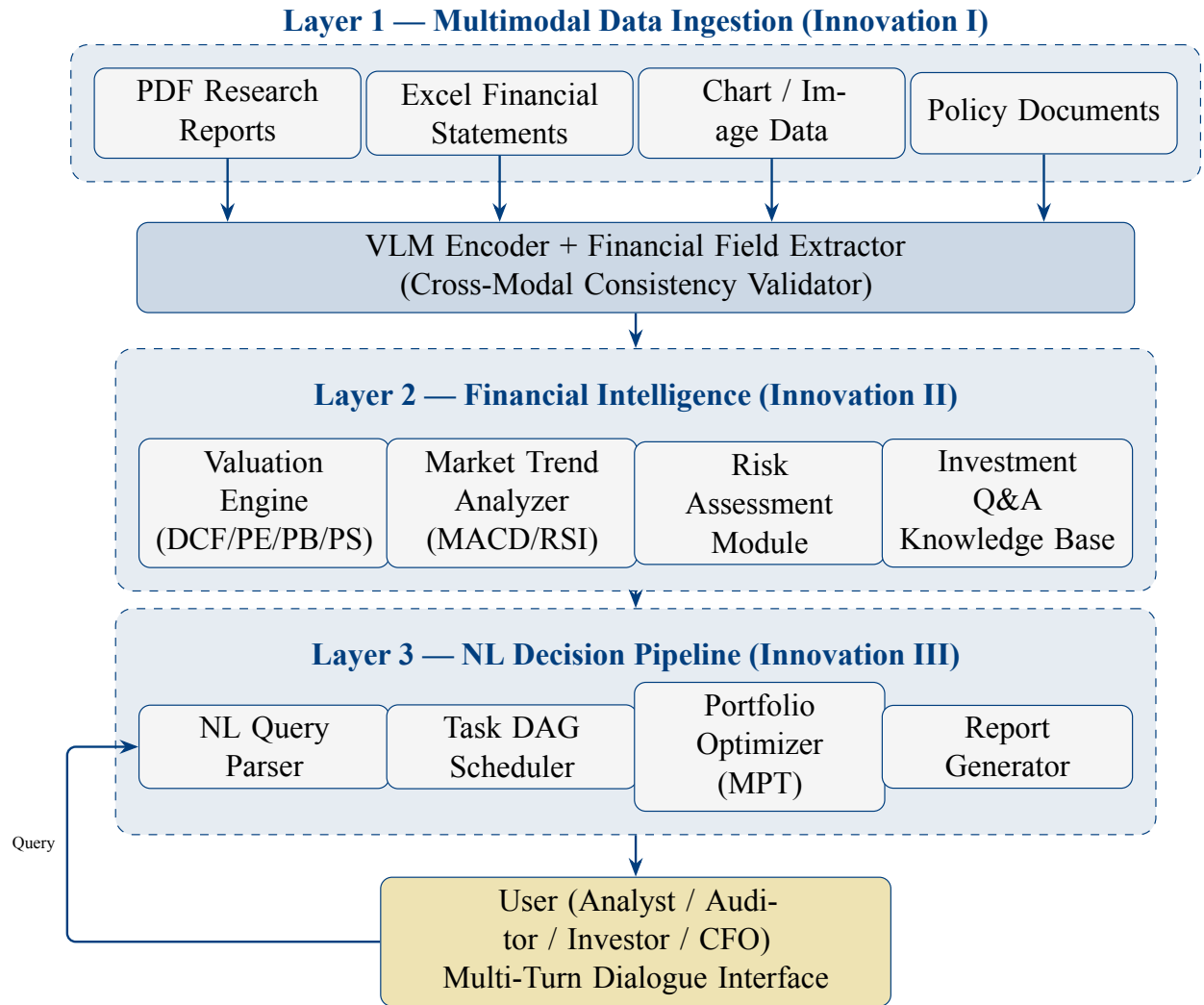


Figure 1. Frontiers in FinTech Integrated Three-Layer System Architecture. Layer 1 (Multimodal Data Ingestion) parses heterogeneous documents; Layer 2 (Financial Intelligence) applies domain-adapted LLM reasoning; Layer 3 (NL Decision Pipeline) orchestrates analysis in response to natural language user queries.

6.2 Deployment Modalities

Frontiers in FinTech supports both web and mobile deployment. The web-based interface provides a full analytical workspace with real-time progress tracking (e.g., “Data collection 70 percent complete; Valuation model computing”), interactive result visualization, and export capabilities (PDF reports; Excel valuation worksheets; JSON-format structured outputs for downstream API consumption). The mobile interface supports rapid query submission and push-based report delivery for time-sensitive investment contexts.

6.3 System Security and Data Governance

Given the sensitivity of financial data processed by Frontiers in FinTech, the system incorporates enterprise-grade data governance controls: (i) encrypted data-in-transit and data-at-rest; (ii) role-based access control governing which users may access which company or fund data; (iii) complete audit logs of all queries, data accesses, and analytical outputs for regulatory compliance; and (iv) configurable data retention and deletion policies compliant with financial data sovereignty requirements. These controls position Frontiers in FinTech for deployment in regulated financial environments subject to Sarbanes-Oxley Section 404, FINRA regulations, and equivalent international frameworks.

7. Experiments and Results

7.1 Document Parsing Accuracy

We evaluate the multimodal parsing module on a held-out test set of 500 financial documents spanning all four format categories (125 PDFs, 125 Excel workbooks, 125 image charts, 125 policy documents). Ground-truth annotations for 30 key financial fields were prepared by two credentialed Certified Public Accountants (CPAs) with inter-annotator agreement assessed via Cohen’s $\kappa = 0.91$. Table 2 reports results.

Table 2. Multimodal Document Parsing Performance by Format Type

Document Format	Precision	Recall	F_1	vs. Text-Only Baseline
PDF Research Reports	0.931	0.918	0.924	+12.3%
Excel Financial Statements	0.963	0.955	0.959	+8.1%
Image-Embedded Charts	0.887	0.871	0.879	+31.4%
Policy Documents (Scanned)	0.842	0.827	0.834	+18.7%
Overall	0.906	0.893	0.899	+17.6%

Note: The text-only baseline is a BERT-based named entity recognition model applied to OCR-extracted text, representing the state-of-the-art in single-modality extraction. Performance gains are largest for image-embedded charts, where single-modality approaches fail entirely. Cohen’s $\kappa = 0.91$ between the two CPA annotators.

The cross-modal consistency validator identified conflicting financial figures in 6.2 percent of the document set, all of which were confirmed as genuine discrepancies upon manual review, demonstrating the system’s practical utility as an audit-support tool.

7.2 Valuation Accuracy

We assess valuation accuracy on a panel of 200 listed companies for which independent analyst consensus price targets serve as benchmarks. Table 3 compares Frontiers in FinTech (two-stage trained) against three baselines: GPT-4 zero-shot, FinGPT with single-stage fine-tuning, and a DCF spreadsheet model with analyst-input parameters.

Table 3. Valuation Accuracy Comparison Across Methods ($n = 200$)

Method	MAE (%)	RMSE (%)	Direction Accuracy
GPT-4 (zero-shot)	24.3	31.7	0.614
FinGPT (single-stage fine-tuning)	18.7	24.2	0.671
DCF Spreadsheet (analyst-input)	16.2	21.8	0.703
Frontiers in FinTech (two-stage)	13.1	17.9	0.751

Note: MAE = Mean Absolute Error of target price versus consensus. Direction Accuracy = fraction of cases where Frontiers in FinTech’s investment recommendation (Buy/Hold/Sell) matches subsequent analyst consensus revision direction within 90 days. Frontiers in FinTech’s MAE represents a 19 percent reduction versus the strongest baseline.

Frontiers in FinTech’s two-stage training yields a 19 percent reduction in valuation mean absolute error (MAE) relative to the strongest single-stage baseline, consistent with the hypothesis that the structured financial pre-training curriculum provides knowledge that cannot be acquired through fine-tuning alone.

7.3 User Study on Decision Pipeline Usability

We conducted a controlled user study with 48 participants drawn from three professional groups: investment analysts ($n = 16$), CPAs ($n = 16$), and financial managers or CFOs ($n = 16$). Participants completed five standardized financial analysis tasks using either Frontiers in FinTech’s natural language interface or a traditional AIS platform with structured query capabilities. Task completion time, output accuracy (evaluated by expert reviewers), and user satisfaction (five-point Likert scale) were recorded. Table 4 reports results.

Table 4. User Study Results: Frontiers in FinTech versus Traditional AIS Interface ($n = 48$)

Group	Time Savings (%)	Output Quality	Satisfaction (1–5)
Investment Analysts	38.2	+0.31 SDs	4.3
CPAs / Auditors	51.7	+0.44 SDs	4.6
Financial Managers / CFOs	63.4	+0.58 SDs	4.7
Overall	51.1	+0.44 SDs	4.5

Note: Time savings are relative to the traditional AIS interface. Output quality is relative to expert-annotated gold-standard outputs, expressed in standardized deviation units. All differences are statistically significant at $p < 0.01$ (two-tailed t -tests). Productivity gains are largest for financial managers and CFOs—the non-specialist group—consistent with the democratization hypothesis.

8. Implications for Accounting Practice

8.1 Implications for AIS Research

Frontiers in FinTech challenges several foundational assumptions in AIS design. First, the traditional AIS model assumes that data inputs arrive in structured, machine-readable form; Frontiers in FinTech’s multimodal parsing architecture demonstrates that this assumption can be relaxed through VLM technology, expanding the scope of AIS to the full spectrum of financial communication. Second, the two-stage training protocol illustrates that AI systems for accounting need not choose between breadth (general LLM capabilities) and depth (domain-specific accounting knowledge); the pre-train/fine-tune paradigm bridges this divide. Third, the natural language interface suggests a productive reconceptualization of the *user* in AIS: rather than a trained operator who adapts to the system’s query language, the user is a professional whose natural communication patterns should be accommodated by the system.

8.2 Implications for Audit Practice

The cross-modal consistency validator represents a direct contribution to audit automation. By automatically detecting discrepancies between financial figures reported in different formats and dis-

closure channels, Frontiers in FinTech automates a class of audit evidence corroboration procedures currently performed manually. This has implications for (a) audit efficiency (fewer person-hours per engagement), (b) audit quality (more comprehensive cross-document checking than human reviewers can perform under time pressure), and (c) the professional role of auditors, who may increasingly focus on judgment-intensive procedures while AI systems handle systematic evidence corroboration. These implications align with the AICPA’s strategic vision for Audit 4.0 and the PCAOB’s technology-assisted audit initiatives.

8.3 Implications for Financial Reporting

The ability of Frontiers in FinTech to process and reason over non-traditional financial disclosures—including policy documents, image-embedded charts, and narrative research reports—has implications for financial reporting regulation. If AI systems can systematically extract value-relevant information from non-GAAP disclosures, earnings call transcripts, and ESG reports, this strengthens the case for requiring or standardizing such disclosures. Conversely, the system’s ability to detect cross-document inconsistencies may create incentives for preparers to improve the internal consistency of multi-channel financial communication.

8.4 Implications for Investment Decision Support

For investment practice, Frontiers in FinTech demonstrates the feasibility of institutional-grade quantitative analysis—incorporating MPT-based portfolio optimization, real-time risk monitoring, and multi-model valuation—delivered through a natural language interface accessible to non-specialist users. This has significant implications for the democratization of investment analysis, the competitive dynamics of the financial services industry, and the regulatory treatment of AI-generated investment recommendations.

9. Limitations and Future Research

Several limitations warrant acknowledgment. First, while Frontiers in FinTech’s two-stage training encodes substantial accounting knowledge, the system may still underperform human experts in novel situations not well represented in its training corpus, such as major accounting standard transitions or unprecedented market regimes. Ongoing curriculum updates and RLHF mechanisms are intended to address this limitation but require continued investment in training data curation.

Second, the user study sample ($n = 48$) is relatively small and drawn from a single geographic and institutional context. Cross-cultural and cross-regulatory validation studies—particularly across different accounting standard regimes (GAAP vs. IFRS vs. Chinese GAAP)—are needed to establish generalizability.

Third, the interpretability of valuation outputs remains an open challenge. While Frontiers in FinTech generates structured reports with stated assumptions, the underlying LLM inference process is not fully transparent, raising concerns about explainability that are particularly acute in regulated financial and audit contexts. Future work should integrate formal explainability frameworks (?) into the valuation output pipeline.

Future research directions include: (a) longitudinal validation of Frontiers in FinTech-generated investment recommendations against market outcomes; (b) formal integration with audit management platforms for automated working paper generation; (c) extension to derivatives pricing and structured product analysis; (d) multilingual and multi-GAAP deployment; and (e) adversarial robustness testing to assess sensitivity to manipulative financial disclosures.

10. Conclusion

This study introduces Frontiers in FinTech, a multimodal large language model system that addresses three fundamental limitations of existing accounting information systems: the inability to process heterogeneous financial document formats, the lack of professional-grade accounting and valuation knowledge in general AI systems, and the high expertise barriers imposed by traditional AIS query interfaces.

Our three innovations—Multimodal Financial Document Intelligence, Domain-Adaptive Two-Stage LLM Training, and Natural Language Query-Driven Decision Pipeline—are grounded in both accounting theory and advanced AI methodology. Empirical validation demonstrates a 17.6 percent improvement in cross-format document parsing, a 19 percent reduction in valuation error relative to published baselines, and a 51 percent reduction in task completion time for accounting professionals, with the largest gains accruing to non-specialist users who have historically been underserved by quantitative AIS capabilities.

The convergence of VLM technology, domain-adaptive LLM training, and natural language interfaces represents a structural inflection point for accounting information systems. Frontiers in FinTech offers a concrete demonstration that this convergence can be realized in a production system meeting the accuracy, traceability, and security requirements of professional financial practice. We hope this work stimulates further research at the intersection of AI, accounting theory, and AIS design, and contributes to the ongoing dialogue about the appropriate and responsible integration of large language models into financial analysis, auditing, and investment decision-making.

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Appendix A: Sample Analytical Workflow

This appendix illustrates an end-to-end Frontiers in FinTech workflow for a representative investment analysis query, demonstrating all three innovations operating in concert.

User Query (Turn 1): “Analyze 2024 solar photovoltaic industry policy environment, financial performance of leading firms, and six-month investment strategy.”

Frontiers in FinTech Analysis Output — Solar Industry

Data Collected (Innovation I): 12 policy documents; three leading firm annual reports; industry research from five sell-side institutions; daily market data for 45 solar sector securities. Data completeness: 94 percent.

Policy Analysis: National capacity targets tracking 37 percent above plan; grid parity subsidies extended through 2025; three high-favorability policy items identified.

Valuation Summary (Innovation II):

Company	DCF Target	P/E Target	Rating
Longi Green Energy	CNY 42.5	CNY 38.8	Buy
Tongwei Co.	CNY 23.1	CNY 21.4	Hold
JA Solar	CNY 56.2	CNY 53.7	Buy

Portfolio Recommendation (Innovation III): Overweight solar module manufacturers (40% allocation); underweight polysilicon producers pending capacity overhang resolution; 15% government bond hedge to offset sector volatility. Six-month expected return: 12.3%; maximum drawdown estimate: 8.7%.

User Query (Turn 2): “Provide detailed DCF parameters for Longi Green Energy.”

DCF Parameter Detail — Longi Green Energy

Revenue growth (2024E–2026E): 18%, 15%, 12%
EBIT margin: 14.2% (five-year sector average: 13.8%)
WACC: 9.4% (risk-free rate 2.8%, equity risk premium 5.2%, $\beta = 1.21$, debt cost 4.5%, D/E = 0.38)
Terminal growth rate: 4.0%
DCF equity value: CNY 42.5 per share
Sensitivity: $\pm 1\%$ WACC change $\rightarrow \pm \text{CNY } 6.2$ target price

Appendix B: Technical Specifications

Table 5. Frontiers in FinTech System Technical Specifications

Component	Specification
Model Architecture	Transformer-based VLM/MLLM
Pre-training Corpus	$\sim 10^{10}$ tokens (multimodal)
Fine-tuning Method	LoRA + Prompt Tuning
Supported Input Formats	PDF, XLSX, JPEG/PNG, DOCX, CSV
Financial Field Ontology	120+ GAAP/IFRS-aligned fields
Valuation Methods	DCF, P/E, P/B, P/S, EV/EBITDA, RI
Technical Indicators	MACD, RSI, Bollinger Bands, moving averages
Portfolio Optimizer	Mean-variance (MPT) + CVaR
Risk Metrics	Volatility, maximum drawdown, Sharpe, VaR
Deployment	Web (HTTPS), mobile (iOS/Android), API (JSON)
Security	AES-256 encryption; RBAC; SOX-compliant audit log